

AI Foundations | AIFo

Summary

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1. MACHINE LEARNING (ML)

There are many different AI research fields and AI applications. Algorithms and applications where a computer learns from data are typically in the field of statistical machine learning (ML).

Computer: performs a mapping from input to output.

Processing: any sort of non-trivial “calculation” (mapping) or information processing

Input: any sort of data/information (sensory input, bits, mechanical configuration...)

Output: any sort of response (data, actions, new state of a system...)

TODO:

W1 slides 2 14..17

2. PROBABILITY

TODO:

merge with MathFML, DigCod, WrStat

2.1. RANDOM VARIABLES

A random variable X is a variable that takes a numerical value x , which depends on a random experiment. Random Variables come in two flavours:

- discrete: X takes any of a finite or countable set of values, e.g. $\{-8, 1.5, 1.5, 2.693, 5, 6.3, 10\}$
- continuous: X takes any value of an uncountable range, e.g. the real numbers in the interval $(2, 7)$.

Before observing the outcome of a random experiment, the ‘best we can know’ about a random variable is a “list” of all possible values, and a second “list” which tells us for each possible value how likely each value will occur.

$\Pr(X = x)$ is the probability that the random variable X takes the value x . It’s often written as $P(x)$ or $p(x)$ or $P_X(x)$.

2.2. PROBABILITY MASS FUNCTION (PMF)

The Probability Mass Function (PMF) of a discrete random variable is a function $f(x)$ that provides the probability for each value x of a discrete random variable X .

Example 1:

Dice rolls

Value x of the random Variable X	1	2	3	4	5	6
$P(X = x)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

TODO:

table s7,8,10

From the PMF, we can calculate the expected value

$$E[X] = \sum_{i=1}^{\infty} P(x_i) \cdot x_i$$

2.2.1. Multiple Random Variables

We can study two random variables simultaneously.

TODO:

slides 12,13

The (joint) properties of two random variables are defined by the **Joint Probability Mass Function**, usually simply called Joint Probability.

For **independent** random variables, the joint probability is simply the product of the individual probabilities

$$P(X, Y) = P(X) \cdot P(Y)$$

Example 2: Rolling two dice

The random experiment of rolling die 1 and die 2 together has 36 possible outcomes (events). The Joint Probability is:

$$P(X = 5, Y = 4) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

	$X = 1$	$X = 2$	$X = 3$	$X = 4$	$X = 5$	$X = 6$
$Y = 1$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
$Y = 2$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
$Y = 3$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
$Y = 4$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
$Y = 5$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
$Y = 6$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

TODO:

For **dependent** random variables,

$$P(Y|X) = \frac{P(X, Y)}{P(X)}$$

slides 17,18,20,22

2.3. PROBABILITY DENSITY FUNCTION (PDF)

$$P(-\infty < x < \infty) = 1$$

$$P(a < x < b) = \int_a^b f(x) dx$$

$$P(x) = 0$$

TODO:

https://en.wikipedia.org/wiki/68%E2%80%939395%E2%80%939399.7_rule